Quantum Information Theory

Fidelity of Quantum States:
An Introduction to Uhlmann's Theorem

Talk 3 — July 9, 2026

Review

Classical discrimination of a classical random variable x drawn from one of the pdf's $p(x)$ or $q(x)$. Rule: given observation x, choose the greater of

$$\begin{cases} p(x) \longrightarrow x \sim p, \\ q(x) \longrightarrow x \sim q, \end{cases} \qquad \Big(\text{gives maximum prob. of success}\Big).$$

Quantum discrimination between two mixed states ρ, σ: we designed the optimal POVM $\{E, I - E\}$. The probability of successful guessing in both cases is

$$P^{\text{succ}}_{\text{classical}} = \frac{1}{2}\left(1 + \frac{1}{2}\left\|p - q\right\|_1\right), \qquad P^{\text{succ}}_{\text{quantum}} = \frac{1}{2}\left(1 + \frac{1}{2}\left\|\rho - \sigma\right\|_1\right),$$

where

$$\left\|A\right\|_1 = \sum_i |\lambda_i(A)| = \operatorname{Tr}\sqrt{A^\dagger A}.$$

We defined the *trace distance* between states ρ and σ,

$$T(\rho, \sigma) = \frac{1}{2}\left\|\rho - \sigma\right\|_1,$$

which is an operational definition in terms of a classification task, and also a nice mathematical definition with an interpretation as a proper metric. Other operational definitions are possible and useful in their own particular applications.

Today we look at fidelity, $F(\rho, \sigma)$, between two states.

1 Fidelity as a Generalization of Overlap

Fidelity is a generalization of the overlap between two pure states, $\langle\psi|\phi\rangle$; we want something similar for mixed states.

Claim. For pure states there is a direct relation between overlap and distinguishability:

$$\langle\psi|\phi\rangle \;\sim\; \frac{1}{2}\left\|\,|\psi\rangle\langle\psi| - |\phi\rangle\langle\phi|\,\right\|_1 \qquad \text{(not equal!)}.$$

Why. Take the 2-dimensional subspace spanned by $|\psi\rangle$ and $|\phi\rangle$. Take basis vector $|e_1\rangle \equiv |\psi\rangle$ and any orthonormal $|e_2\rangle$. Then

$$|\psi\rangle = \begin{pmatrix}1\\0\end{pmatrix}, \qquad |\phi\rangle = \cos\theta\ |e_1\rangle + \sin\theta\ |e_2\rangle = \begin{pmatrix}\cos\theta\\ \sin\theta\end{pmatrix}, \quad \text{where } \langle\psi|\phi\rangle = \cos\theta.$$

So, writing $c = \cos\theta$ and $s = \sin\theta$,

$$\begin{aligned}|\psi\rangle\langle\psi| - |\phi\rangle\langle\phi| &= \begin{pmatrix}1\\0\end{pmatrix}\begin{pmatrix}1 & 0\end{pmatrix} - \begin{pmatrix}c\\s\end{pmatrix}\begin{pmatrix}c & s\end{pmatrix} \\ &= \begin{pmatrix}1 & 0\\ 0 & 0\end{pmatrix} - \begin{pmatrix}c^2 & cs\\ cs & s^2\end{pmatrix} = \begin{pmatrix}s^2 & -cs\\ -cs & -s^2\end{pmatrix},\end{aligned}$$

and we need $\sum_{i=1}^{2} |\lambda_i|$ of its eigenvalues.

Recall the characteristic equation for a 2×2 matrix, $\lambda^2 - \mathrm{Tr}(A)\,\lambda + \det(A) = 0$. Here $\mathrm{Tr} = 0$ and

$$\det = -s^4 - c^2 s^2 = -s^2(s^2 + c^2) = -\sin^2\theta,$$

so

$$\lambda^2 - \sin^2\theta = 0 \quad \Longrightarrow \quad \lambda = \pm\sin\theta.$$

Therefore $|\lambda_1| + |\lambda_2| = 2\sin\theta$, and

$$\boxed{\frac{1}{2}\,\|\,|\psi\rangle\langle\psi| - |\phi\rangle\langle\phi|\,\|_1 = \sin\theta = T(|\psi\rangle, |\phi\rangle), \qquad \text{but} \quad |\langle\psi|\phi\rangle|^2 = \cos^2\theta.}$$

This is a simple 1:1 relation for pure states. This *squared-overlap* is what we call fidelity for pure states.

2 General Definition of Fidelity

For two mixed states ρ, σ in general,

$$\boxed{F(\rho, \sigma) = \left\|\sqrt{\rho}\,\sqrt{\sigma}\right\|_1^2.}$$

2.1 Recovering the pure-state definition

Claim. This recovers the pure-state definition above: $\cos^2\theta$.

Proof. Take $\rho = |\psi\rangle\langle\psi|$ and $\sigma = |\phi\rangle\langle\phi|$; we want to show $\left\|\sqrt{\rho}\,\sqrt{\sigma}\right\|_1^2 = |\langle\psi|\phi\rangle|^2$. Note that for a rank-arbitrary projection such as $\Pi = |\psi\rangle\langle\psi|$ we have $\Pi^2 = \Pi$, hence $\Pi = \sqrt{\Pi}$. So

$$\begin{aligned}F(\psi, \phi) &= \left\|\sqrt{|\psi\rangle\langle\psi|}\,\sqrt{|\phi\rangle\langle\phi|}\right\|_1^2 = \|\,|\psi\rangle\langle\psi|\ |\phi\rangle\langle\phi|\,\|_1^2 \\ &= \left\|\underbrace{\langle\psi|\phi\rangle}_{\in\mathbb{C}}\ |\psi\rangle\langle\phi|\right\|_1^2 = |\langle\psi|\phi\rangle|^2\ \|\,|\psi\rangle\langle\phi|\,\|_1^2,\end{aligned}$$

using $\|cA\|_1 = |c|\ \|A\|_1$. It remains to show the last one-norm equals 1.

One-norm of the rank-one operator $|\psi\rangle\langle\phi|$. Take $X = |\psi\rangle\langle\phi|$, so $\|X\|_1 = \operatorname{Tr}\sqrt{X^\dagger X}$ from the definition. Then

$$X^\dagger X = |\phi\rangle\underbrace{\langle\psi|\psi\rangle}_{=1}\langle\phi| = |\phi\rangle\langle\phi|\,,$$

and since $|\phi\rangle\langle\phi|$ is a rank-one projection, $\sqrt{X^\dagger X} = |\phi\rangle\langle\phi|$. Therefore

$$\operatorname{Tr}\sqrt{X^\dagger X} = \operatorname{Tr}|\phi\rangle\langle\phi| = \operatorname{Tr}\langle\phi|\phi\rangle = \operatorname{Tr}1 = 1 \quad \Longrightarrow \quad \|\,|\psi\rangle\langle\phi|\,\|_1 = 1.$$

Hence

$$\boxed{F(|\psi\rangle\,, |\phi\rangle) = |\langle\psi|\phi\rangle|^2,}$$

as expected. □

2.2 Fidelity of a pure and a mixed state

Claim. $F(|\psi\rangle\,, \rho) = \langle\psi|\rho|\psi\rangle$.

Proof. With $\Pi = |\psi\rangle\langle\psi|$,

$$F(\psi, \rho) = \left\|\sqrt{|\psi\rangle\langle\psi|}\,\sqrt{\rho}\right\|_1^2 = \|\,|\psi\rangle\langle\psi|\ \sqrt{\rho}\,\|_1^2 = \|\Pi\sqrt{\rho}\|_1^2 \qquad (\Pi \text{ a projection}).$$

Using $\|A\|_1 = \operatorname{Tr}\sqrt{AA^\dagger}$,

$$\|\Pi\sqrt{\rho}\|_1 = \operatorname{Tr}\sqrt{\Pi\sqrt{\rho}\,\sqrt{\rho}\,\Pi} = \operatorname{Tr}\sqrt{\Pi\rho\Pi}.$$

Now

$$\Pi\rho\Pi = |\psi\rangle\langle\psi|\,\rho\,|\psi\rangle\langle\psi| = \underbrace{\langle\psi|\rho|\psi\rangle}_{\in\mathbb{C}}\ |\psi\rangle\langle\psi|\,,$$

so

$$\operatorname{Tr}\sqrt{\Pi\rho\Pi} = \sqrt{\langle\psi|\rho|\psi\rangle}\ \operatorname{Tr}\sqrt{|\psi\rangle\langle\psi|} = \sqrt{\langle\psi|\rho|\psi\rangle}\ \underbrace{\operatorname{Tr}|\psi\rangle\langle\psi|}_{=1} = \sqrt{\langle\psi|\rho|\psi\rangle}.$$

Thus $\left\|\Pi\sqrt{\rho}\right\|_1 = \sqrt{\langle\psi|\rho|\psi\rangle}$, and

$$\boxed{F(\psi, \rho) = \langle\psi|\rho|\psi\rangle\,.}$$

□

2.3 Operational interpretation

Both cases have a similar operational interpretation as the probability of a measurement outcome for the POVM $\{|\psi\rangle\langle\psi|\,,\ I - |\psi\rangle\langle\psi|\}$:

$$\begin{aligned} F(\psi, \phi) = |\langle\psi|\phi\rangle|^2 \quad &= \text{prob. of outcome } \psi \text{ when given state } |\phi\rangle\,; \\ F(\psi, \rho) = \langle\psi|\rho|\psi\rangle \quad &= \text{prob. of outcome } \psi \text{ when given state } \rho. \end{aligned}$$

3 Mixed States

For two mixed states,

$$F(\rho, \sigma) = \left\|\sqrt{\rho}\,\sqrt{\sigma}\right\|_1^2\,.$$

3.1 Commuting operators

For commuting operators it is easy to show

$$F(\rho,\sigma) = \left(\sum_i \sqrt{p_i\, q_i}\right)^2 ,$$

where p_i are the eigenvalues of ρ and q_i are the eigenvalues of σ. This is the Bhattacharyya distance of the classical probability distributions $\{p_i\}, \{q_i\}$.

3.2 Non-commuting mixed states

How do we measure an "overlap" of mixed states? Recall the *purification* of mixed states. Take a mixed state ρ_A on a system A; add a reference system R and produce a *pure state* $|\psi\rangle_{RA}$ on the joint system RA such that discarding (tracing out) R on the joint state produces ρ_A:

$$\rho_A \xrightarrow{\text{purification}} |\psi\rangle_{RA} \qquad \text{s.t.} \qquad \mathrm{Tr}_R\, |\psi\rangle_{RA}\langle\psi|_{RA} = \rho_A.$$

Do this purification on both ρ and σ, giving $\rho \to |\psi_\rho\rangle$ and $\sigma \to |\psi_\sigma\rangle$. Try

$$F(\sigma_A, \rho_A) \stackrel{?}{=} |\langle\psi_\rho|\psi_\sigma\rangle|^2.$$

Not quite! It turns out this is *not unique* — but there *is* a unique maximum. So

$$\boxed{F(\rho_A, \sigma_A) = \left\|\sqrt{\rho}\,\sqrt{\sigma}\right\|_1^2 \quad \text{(by definition)} \qquad \text{ALSO} \qquad = \max_{\text{purifications}} |\langle\psi_\rho|\psi_\sigma\rangle|^2.}$$

This is Uhlmann's Theorem.

4 Review of Purification

Recall: (1) a pure state $|\psi\rangle$; and (2) a mixed state ρ with $\rho \geq 0$ and $\mathrm{Tr}\,\rho = 1$.

We can view a mixed state ρ_A on a Hilbert space A as arising from a pure state $|\psi_\rho\rangle_{RA}$ on an extended Hilbert space $R \otimes A$,

$$\rho_{RA} = |\psi_\rho\rangle_{RA}\langle\psi_\rho|_{RA}\,,$$

then discarding system R and keeping only A. Here $|\psi\rangle_{RA}$ is the *purification* of ρ_A. This is *not unique*: infinitely many different $|\psi_\rho\rangle_{RA}$ give the same ρ_A after discarding R.

4.1 Construction

Given ρ_A, take its eigendecomposition $\rho_A = \sum_i p_i\, |e_i\rangle\langle e_i|$, and define

$$\boxed{|\psi_\rho\rangle_{RA} = \sum_i \sqrt{p_i}\; |i\rangle_R\; |e_i\rangle_A\,,}$$

where $|i\rangle_R$ is an arbitrary orthonormal basis in another Hilbert space with $\dim(R) = \dim(A)$.

Check. Trace out system R by applying $\mathrm{Tr}_R \otimes I_A$ (trace on R, identity on A) to $|\psi_\rho\rangle_{RA}\langle\psi_\rho|_{RA}$:

$$\begin{aligned}\mathrm{Tr}_R \sum_{i,j} \sqrt{p_i p_j}\ |i\rangle\langle j|_R \otimes |e_i\rangle\langle e_j| &= \sum_{i,j} \sqrt{p_i p_j}\ \underbrace{\mathrm{Tr}\big(\,|i\rangle\langle j|_R\,\big)}_{=\delta_{ij}} \otimes |e_i\rangle\langle e_j| \\ &= \sum_i p_i\ |e_i\rangle\langle e_i| = \rho_A. \qquad \checkmark\end{aligned}$$

4.2 Non-uniqueness and Uhlmann's theorem

Purifications are *not unique.* In

$$|\psi_\rho\rangle_{RA} = \sum_i \sqrt{p_i}\ |i\rangle_R \otimes |e_i\rangle_A\,,$$

the $|e_i\rangle$ form an ONB on A (eigenvectors of ρ_A) while $|i\rangle_R$ is an arbitrary ONB on another Hilbert space. We are free to rotate the ONB on R with a unitary U_R: that is, $(U_R \otimes I_A)\,|\psi_\rho\rangle_{RA}$ is also a purification for ρ_A, given a purification $|\psi_\rho\rangle_{RA}$.

Relation to fidelity between states ρ and σ. Given purifications $\rho_A \to |\psi_\rho\rangle_{RA}$ and $\sigma_A \to |\psi_\sigma\rangle_{RA}$, we can define *a* fidelity between ρ and σ as the overlap of the purifications, $|\langle\psi_\rho|\psi_\sigma\rangle|^2$. Define *the* fidelity between ρ and σ as

$$\boxed{F(\rho,\sigma) = \max_{\text{purifications}} |\langle\psi_\rho|\psi_\sigma\rangle|^2.}$$

One then shows this also equals $\big\|\sqrt{\rho}\,\sqrt{\sigma}\big\|_1^2$ — with no optimization procedure required, via a direct calculation.

This is Uhlmann's Theorem.